Local Income Taxes and Residential Sorting:

Evidence from Pennsylvania LEHD Origin-Destination Data

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Abstract

Do local income tax rate changes cause residential sorting across municipal boundaries? Using LEHD Origin-Destination data linking each worker's residence to their workplace at municipal resolution, I estimate the effect of Pennsylvania municipal Earned Income Tax (EIT) rate changes on the locally-resident share and resident worker count of each city's workforce, 2002–2019. Pennsylvania's universal EIT, with continuous rate variation across ∼2,500 municipalities and ∼500 school districts, and Act 32 of 2008 (effective January 1, 2012) consolidating collection and mandating employer withholding, provides the cleanest sub-state setting for identifying local-income-tax sorting in the United States. A 1 percentage-point increase in the muni resident rate lowers the live-and-work-in-city share by 0.8 share-points and resident worker counts by 11 percent. Restricting to a clean sample (muni-rate increasers vs. never-changers, dropping flip-floppers), the muni-rate effect rises to −2.0 share-points and −26 percent on resident workers per pp; decreases produce statistically imprecise null estimates. The response is concentrated entirely after Act 32: the muni rate coefficient on `share_live_in` is essentially zero pre-2012 and −1.1 to −1.5 share-points per pp post-2012, confirming that residents respond to taxes that are actually collected. School

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district rates have the opposite sign: a 1 pp SD rate increase raises resident workers by 14 percent, controlling for muni rate. Bundling muni and SD rates into a single "total EIT" rate hides both channels; decomposition is essential. Rate increases are fiscally productive in the magnitudes we estimate: behavioral responses are too small to erode the tax base.

1 Introduction

The Tiebout hypothesis—that households "vote with their feet" by sorting across jurisdictions offering different bundles of taxes and public services (Tiebout, 1956)—remains a central organizing principle in local public finance. A large empirical literature documents tax-driven migration at the state level for high earners (Kleven et al., 2020; Young et al., 2016; Moretti and Wilson, 2017) and at sub-national borders for sales taxes (Agrawal, 2015). Yet sub-state income-tax sorting—the canonical Tiebout margin, where moving costs are small and tax boundaries are crossed by a short relocation—is essentially untested in the United States. Two recent studies of Indiana and Maryland local income taxes (Yang and Heim, 2025; Heim, 2017) use county-level IRS data and find county-level migration responses but cannot resolve the muni-level variation that defines the actual tax boundary. A literature survey by Agrawal (2025) explicitly identifies sub-state local-income-tax sorting as an open question, and Agrawal et al. (2022) characterize the empirical evidence as "thin."

The gap persists for two reasons. First, the jurisdictions that levy local income taxes are sub-county, so standard migration data sources do not see them. The IRS Statistics of Income tracks county-to-county migration; the American Community Survey measures place-level population but does not link workers to employers. Second, local income-tax systems feature credit mechanisms, overlapping school-district taxes, and uneven enforcement that complicate the mapping from statutory rates to effective tax burden. A naive comparison of rates and population changes conflates tax incidence with amenity valuation, fiscal capacity, and enforcement quality.

This paper addresses both challenges using LEHD Origin-Destination Employment Statistics (LODES), the only administrative data source that links each worker's residence tract to their workplace tract at annual frequency. The setting is Pennsylvania, 2002–2019. Pennsylvania operates a universal Earned Income Tax under Act 511 of 1965 with continuous rate variation across approximately 2,500 municipalities and 500 school districts, and a quasi-experimental enforcement break: Act 32 of 2008 (effective January 1, 2012) consolidated

EIT collection into 69 Tax Collection Districts and mandated employer withholding at the higher of the home resident rate or the work nonresident rate. Before Act 32, withholding was fragmented and many municipalities did not even publish a nonresident rate. The structural break in enforcement—same statutory rates, very different effective tax burden—anchors causal identification.

The headline finding is that a 1 percentage-point increase in the muni resident rate reduces the share of workers living and working in the city by 0.8 share-points and the count of resident workers by 11 percent (with full controls and pair fixed effects). Restricting to a clean sample of muni-rate increasers versus never-changers (dropping flip-floppers), the per-pp effect rises to -2.0 share-points and -26 percent on resident workers. Decreases produce statistically imprecise null estimates, consistent with sticky moving costs. Decomposing the home-side rate into the muni resident component and the school-district component reveals an opposite-sign pattern: SD rate increases raise resident worker counts by about 14 percent per pp, controlling for muni rate. Bundling muni and SD rates into a single "total EIT" regressor obscures both channels; the decomposition is essential.

The response is concentrated entirely after Act 32. The muni rate coefficient on `share_live_in` is statistically zero pre-2012 and -1.1 to -1.5 share-points per pp post-2012, confirming that residents respond to taxes that are actually collected. The same pattern holds across the staggered-DiD-robust Sun-Abraham estimator (Sun and Abraham, 2021) and the continuous-treatment estimator of de Chaisemartin et al. (2024), which gives an average causal response of -0.9 share-points per pp with clean pre-trends.

The findings have implications for both methodology and policy. The methodological contribution is that local-income-tax variation is identifiable at the muni level using LEHD data, complementing rather than replicating the county-level results in Yang and Heim (2025) and the income-margin results in Heim (2017). The bias from naive analyses that aggregate rates is documented: the total-stack EIT rate produces null estimates even when the muni and SD components individually move outcomes in opposite directions. Substantively, the

response magnitudes are too small to erode the tax base—a 1 pp muni rate increase reduces resident workers by about 0.3 percent of the muni population in the average city, leaving fiscal capacity essentially intact.

2 Institutional Background

Pennsylvania's Earned Income Tax (EIT) is the largest source of local-income-tax variation in the United States. Approximately 2,500 municipalities and 500 school districts each set their own rate, covering essentially all earned income earned in or by Pennsylvania residents. Unlike property taxes, which are paid by owners and capitalized into home values, the EIT falls directly on workers and varies across municipal boundaries within the same metropolitan area. This creates sharp spatial discontinuities in tax burden that can influence where workers choose to live and work.

2.1 Pennsylvania Earned Income Tax

Pennsylvania's EIT is mandatory for all municipalities under the Local Tax Enabling Act (Act 511 of 1965). It applies to all earned income with no minimum threshold; retirees are exempt. The standard combined cap is 1%, typically split as 0.5% municipal and 0.5% school district. Home Rule municipalities may exceed this cap, and Philadelphia operates separately under the Sterling Act of 1932 at approximately 3.9%. The EIT is the primary revenue lever for most Pennsylvania municipalities: property tax millage caps are binding in many jurisdictions. Our panel (2002–2019) contains 210 muni rate increases, 119 muni rate decreases, 611 school-district rate increases, and 52 school-district rate decreases.

Three rates per municipality. Each Pennsylvania municipality has three EIT rates relevant to a worker:

- $\tau^{\text{res,muni}}$, the muni resident rate, levied on residents of the muni;

- $\tau^{\text{nr,muni}}$, the muni nonresident rate, levied on workers employed in the muni who live elsewhere;
- τ^{SD}, the school-district rate, levied on residents only (school districts cannot tax nonresidents).

Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (muni, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

The piecewise tax allocation. Under Act 32 (effective January 1, 2012), a worker's total EIT bill is determined by the maximum of the home-side rate stack and the work-location nonresident rate, and the bill is split piecewise between home and work jurisdictions:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr,muni}},\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr,muni}} \\ \left(0,\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr,muni}} \end{cases} \tag{1}$$

where j is the worker's home muni and i is the work muni. The work jurisdiction always collects its full nonresident rate first; the home muni and school district then split whatever portion of the worker's bill is left over after the work claim. If the work nonresident rate exceeds the home stack, the home jurisdictions get nothing. The worker's total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max\left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}}\right). \tag{2}$$

Implications for sorting. Equation (1) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{nr,muni}}$), the worker can lower the bill only by moving home to a lower-rate muni or school district. When the work nonresident rate binds, the worker can lower the bill only by changing job to a lower-nonres-rate work

location; moving home without changing job has no effect. For workers who live and work in the same muni, only the home stack applies—they pay $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$.

A worker who lives in muni j and moves to a lower-rate muni j' while keeping the same job at i saves at least the SD rate at j: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which is at most the old bill τ_j^{home} minus τ_j^{SD}. The often-stated claim that "moving residence under a credit system yields zero tax savings" is wrong under Pennsylvania's piecewise structure—residence moves do save tax, because the home-side stack includes the SD rate that does not reappear at the work side.

Philadelphia (excluded from the Act 32 framework). Philadelphia operates under the Sterling Act of 1932 rather than Act 32. Philadelphia residents pay the city resident rate regardless of work location with no offset; Philadelphia nonresidents pay the city nonresident rate, with no residual flowing to home jurisdictions in other Pennsylvania munis. Our main estimates retain Philadelphia in the sample for completeness; results are robust to dropping it.

Act 32: Tax Collection Consolidation. Before 2012, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Nonresident EIT enforcement was weak because no centralized system linked employers, workers, and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The piecewise rule described in Equation 1 existed on paper but was not effectively enforced.

Act 32 of 2008 (effective January 1, 2012) transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *binding*: rate changes now mechanically appear in every worker's paycheck through

employer withholding, regardless of whether they reside in the taxing municipality. Second, it eliminated the possibility of de facto tax avoidance through residential relocation. Before Act 32, moving away from a rate-raising city could effectively reduce a worker's tax burden because the work municipality lacked the enforcement capacity to collect from nonresidents. After Act 32, the credit system is fully enforced, and moving away while keeping the same job provides no tax relief.

School District EIT. School districts in Pennsylvania also levy an EIT under Act 511, typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across our panel is -0.12. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Conceptual Framework

The piecewise tax allocation in Equation (1) and the Act 32 enforcement break together generate testable predictions about *who* responds, *when*, and *which rate* matters. Pennsylvania's institutional structure—two home-side rates (muni resident and school district) plus one work-side rate (muni nonresident), all set independently—makes the response margin diagnostic.

3.1 Three Channels of Adjustment

When a municipality A raises its EIT rate, residents of A can respond through three channels, each with different implications for their tax burden:

Channel 1: Residential relocation without job change. A worker moves residence from A to a lower-rate muni B while continuing to work at A. Under the piecewise allocation, the new bill is $\max(\tau_B^{\text{home}}, \tau_A^{\text{nr,muni}})$, which is at most $\tau_A^{\text{home}} - \tau_A^{\text{SD}}$. So the worker saves *at least* the home school-district rate at A, and typically more if $\tau_B^{\text{home}} < \tau_A^{\text{nr,muni}}$ (work binds at the new home location). Moving residence under Pennsylvania's piecewise EIT *does* produce real tax savings, contrary to the simple credit-system framing in much of the literature.

Channel 2: Complete relocation (residence and job). A worker moves both residence and job from A to B, where $\tau_B^{\text{home}} < \tau_A^{\text{home}}$. Their new liability is τ_B^{home}, yielding the largest savings. This is the canonical Tiebout sorting response.

Channel 3: Compositional change in new hires. When A creates new positions or replaces departing workers, new hires may choose to live outside A given the higher rate. The new hire's bill depends on which side binds, but the choice between candidate home munis favors low-rate ones for a given work location. This channel reduces A's share_live_in even when no existing resident moves.

3.2 Predictions

The institutional features generate three central predictions:

Prediction 1: Enforcement matters (Act 32 break). If sorting is driven by workers optimizing their tax burden, the effect should depend on whether the tax is actually collected. Before Act 32, withholding was fragmented and many munis did not even publish a nonresident rate—55% reporting in 2002, climbing to 60% by 2011. Coverage jumped to 72% in 2012 and plateaued at 77% by 2019. This is not a regime change in statutory rates but in enforcement. The sorting response should therefore be a precise zero pre-2012 and significant post-2012.

Prediction 2: Asymmetry under sticky moving costs. Residential relocation has fixed costs (search, transaction, social network). When A raises its rate, some residents leave once the expected lifetime saving exceeds the fixed cost. When A later cuts its rate, the displaced residents do not move back (the fixed cost is sunk and they have established networks elsewhere). The response should therefore be asymmetric: $|\beta^+| > |\beta^-|$ in cumulative positive vs. negative rate change specs, or analogously in clean-group inc-only vs. dec-only comparisons.

Prediction 3: Muni and SD rates have opposite signs. Muni rate increases are typically fiscal-need motivated (police, roads, debt service). School-district rate increases under Act 1 of 2006 were authorized specifically to fund property-tax relief and school spending. Higher SD rates may therefore signal a *valued amenity* (better schools, potentially lower property taxes) rather than extractive taxation. If so, SD rate increases should *attract* residents, offsetting the repulsive effect of muni rate increases. The two channels are separately identifiable because muni and SD rates vary nearly independently across the panel ($r = -0.12$).

4 Data

4.1 Data Construction Pipeline

The analysis links two administrative systems that do not share common identifiers: LEHD/LODES employment records (Census block geography) and Pennsylvania EIT rates (DCED municipality IDs). The analysis panels are constructed through the following steps:

1. **Scrape PA DCED MunStats tax rates.** PA does not publish a downloadable panel of historical EIT rates. The only source is the DCED MunStats portal, which serves one XLS file ($\sim$12 MB) per year via an ASP.NET form. A Python script submits requests for each year 2000–2026, extracting resident and nonresident EIT rates, school

district EIT rates, and Local Services Tax for all ∼2,500 municipalities. About 27% of munis report multiple school-district rates because their territory spans multiple school districts; in the muni-level panel we assign the maximum SD rate per muni-year.

2. **Scrape PA DCED Annual Financial Reports.** Fiscal controls and revenue shares are constructed from the DCED Annual Financial Reports (71 columns per municipality-year) for 2002–2023, trimmed at load to 2002–2019. Variables include log spending on police, streets, debt service, and intergovernmental state aid, plus revenue shares (EIT, real estate tax, intergov as fractions of total revenue).

3. **Backfill nonresident rates.** The DCED nonresident registry was incomplete pre-Act 32: only 55–60% of munis reported a nonresident rate in 2002–2011. Coverage jumped to 72% in 2012 (Act 32 effective date) and plateaued near 77% by 2019. Diagnostics show this is a reporting gap, not a regime gap: only 0.2% of late-reporting munis have a true zero nonresident rate, and 96.3% of nonresident rates are perfectly stable across 2012–2019. We backfill missing pre-2012 nonresident rates with the earliest observed post-2012 rate per muni and flag the imputation.

4. **Download LODES origin-destination files at the block level.** LODES Version 8 (2020 Census geography throughout, retroactively applied to 2002–2021 by LEHD) is downloaded at the Census block level (`agg_geo = "block"` in `lehdr`). We download main files (intra-state) and auxiliary files (cross-state, where the workplace is in-state but the residence is out-of-state) for PA, 2002–2021. Auxiliary files are necessary because excluding cross-state commuters would mechanically inflate share_live_in for border munis. Segments include all jobs (S000), earnings categories (SE01, SE03), and age groups (SA01–SA03). Reference date: April 1 of each year.

5. **Crosswalk PA DCED to Census geography.** PA rates use DCED municipality IDs with no Census equivalent. PA municipalities are county subdivisions (COUSUBFP), not Census places. We match DCED municipality names to TIGER/Line COUSUBFP

codes using normalized name + county. Match rate: 99.6%. The 9 unmatched munis are boroughs and townships merged or dissolved before the 2020 Census.

6. **Build a block-to-cousub crosswalk.** LODES geographies (block GEOIDs) do not directly identify the cousub. We construct a block-to-cousub crosswalk from the TIGER 2020 PA blocks shapefile by point-in-polygon (block centroid `st_within` cousub). Because 2020 Census blocks nest within cousubs by construction, this assignment is exact—every block has an unambiguous cousub. The crosswalk recovers all 2,561 PA cousubs. We adopt block-level aggregation rather than the tract-level alternative because in rural PA a single tract often spans multiple cousubs; assigning each tract to one cousub by tract centroid silently drops 1,082 of 2,561 cousubs (42%) and misattributes employment for the cousubs it does retain.

7. **Compute neighbor pairs.** Centroid-to-centroid distances between all PA muni pairs; retain pairs within 80 km (the plausible commuting and relocation zone).

8. **Aggregate block-level LODES to municipality-level OD panels.** Raw block-level LODES has ∼5 million block-pair-year observations for PA per year. Each block-pair is mapped to a (work cousub, home cousub) pair via the block-to-cousub crosswalk and summed to produce, for each (work muni, year): total jobs, live-and-work (workers who reside in the work muni), commuters from other in-state munis, and cross-state commuters. This produces `share_live_in`, the primary outcome: the fraction of a city's workforce that also lives there.

9. **Merge rates and controls.** Final step joins rates and fiscal controls to the OD panel, producing the analysis-ready municipality × year dataset.

4.2 Data Sources

LODES Origin-Destination Data. LEHD LODES Version 8, 2002–2019. Block-level residence–workplace links derived from state Unemployment Insurance wage records and the

Quarterly Census of Employment and Wages (Abowd et al., 2009). We trim 2020–2021 due to COVID-related disruptions in both employment patterns and LODES coverage. The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality; the mechanism is calibrated so that aggregation to larger geographies produces estimates that approximate ground truth and have been validated against external sources at the tract level and above (Machanavajjhala et al., 2008; Graham et al., 2014; US Census Bureau, 2023). We aggregate blocks to county subdivisions (cousubs) using the TIGER 2020 block-to-cousub crosswalk described in Section 4.1, item 3.

Municipal Tax Rate Panel. Pennsylvania: DCED MunStats, scraped annually. Resident and nonresident municipal EIT rates, school district EIT rates for all ∼2,500 municipalities. Name-based crosswalk to Census COUSUBFP (99.6% match rate). Pre-2012 nonresident rates backfilled from earliest observed post-2012 value per muni (the registry was incomplete pre-Act 32).

Municipal Financial Reports. PA DCED Annual Financial Reports, 2002–2019 (trimmed from 2002–2023 to exclude COVID/ARPA fiscal distortion). 71 columns per municipality-year including EIT revenue, real estate tax revenue, police expenditure, streets expenditure, total debt, state intergovernmental aid, population, and assessed market value of real estate.

Crime. FBI Uniform Crime Reports, Offenses Known and Clearances by Arrest (Jacob Kaplan concatenated files), 2002–2019. Monthly agency-level counts of Part I index crimes (murder, robbery, aggravated assault, burglary, larceny-theft, motor vehicle theft). Aggregated to annual totals per agency. Matched to PA municipalities via `fips_place_code` (93.5% agency match rate).

Building Permits. Census Bureau Building Permits Survey (BPS), place-level annual files, 2002–2019. Total housing units permitted by structure type (1-unit, 2-unit, 3–4 unit,

5+ unit). PA municipalities matched via `fips_mcd_code` = COUSUBFP (93.3% MCD match rate).

4.3 Sample Restrictions and Transformations

1. **Sample period: 2002–2019.** LODES coverage begins in 2002. We trim at 2019 because COVID disrupted both commuting patterns (LODES 2020–2021) and municipal finances (ARPA transfers distort AFR data 2020–2023). Including these years would conflate pandemic-driven relocation with tax-motivated sorting.

2. **Minimum employment threshold: 50 total jobs.** Small municipalities have share_live_in values dominated by sampling noise (a single worker moving changes the ratio by $1/N$). The 50-job threshold balances precision against sample size. Sensitivity analysis (Section 7) confirms stability across thresholds from 10 to 500.

3. **Rate outlier filters.** PA DCED reports the Local Services Tax (a flat dollar amount) in the same field as the EIT rate (a percentage) in some years. Dropping rates $>$ 5% removes these data entry errors without affecting any real EIT rate (the highest standard EIT is $\sim$1%; only Philadelphia exceeds 3%).

4. **April 1 alignment.** LODES records employment as of April 1. A rate change that takes effect mid-year could be misattributed if not aligned. We assign each municipality-year the rate in effect on April 1, ensuring the tax rate a worker faces and the OD flow recording their location refer to the same date.

5. **Outcome variables.** The primary outcome is $\text{share_live_in}_{it} = \text{live_and_work}_{it}/\text{total_jobs}_{it}$, the locally-resident fraction of a city's workforce. Unlike population counts (which mix retirees, students, and non-workers) or IRS migration data (which is county-level), share_live_in isolates the residential choices of active workers at municipal resolution.

   To decompose the share response into levels, we also estimate effects on log(resident

workers) and log(total jobs) separately. Fiscal outcomes include log EIT revenue, log total tax revenue, log total revenue, log police expenditure, log population, and log market value of real estate from the PA DCED Annual Financial Reports. Pre-treatment balance tests cover eleven observables including crime and building permits.

6. **Treatment definitions.** We use both a continuous and a binary treatment definition. *Continuous*: the current rate level (resident municipal, school district, and the home stack as defined in Equation 1), capturing the intensive margin via TWFE and DCDH. *Binary cohorts*: 1 after a municipality's first rate increase (or first decrease), enabling Sun-Abraham estimation of staggered-DiD-robust dynamic effects.

7. **Control group.** Pennsylvania's EIT is universal, so the relevant control group is municipalities with a stable rate over 2002–2019 (1,264 never-changed municipalities). For the Q3 clean-groups specification we further partition treated municipalities into increase-only (97) and decrease-only (24), dropping the flip-floppers that change in both directions.

8. **Fiscal controls.** Rate increases may coincide with other fiscal changes (police hiring, debt issuance) that independently affect residential attractiveness. We include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid as time-varying controls, plus revenue shares (EIT, real-estate, intergovernmental) from the PA DCED AFR. The main estimates are unchanged with and without these controls (Table 5).

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Municipality × years	26,650 (1,479 municipalities)				

Notes: Pennsylvania municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. Municipalities are Census county subdivisions.

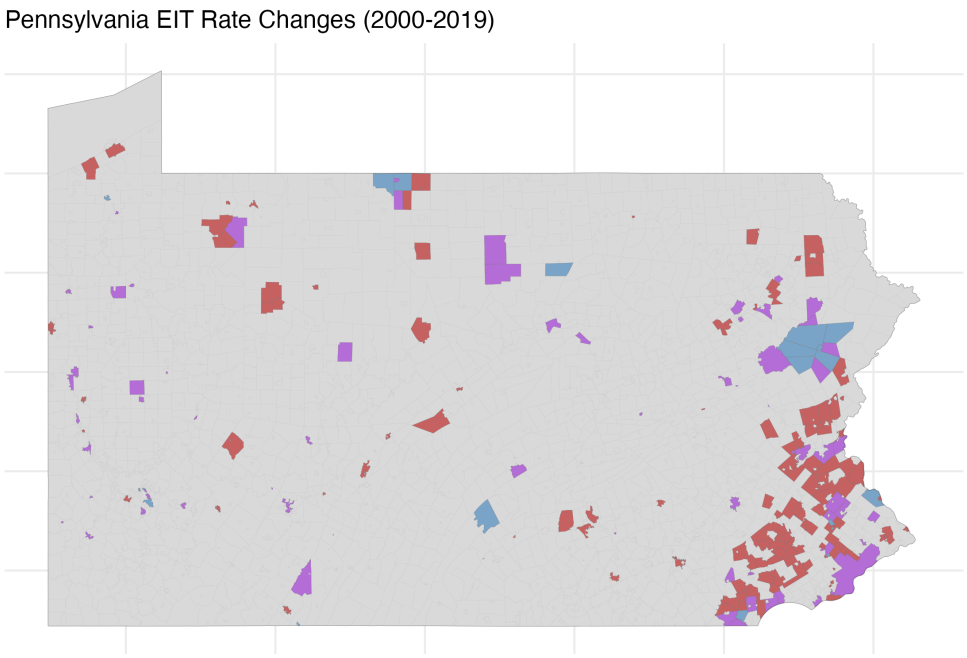


Figure 1: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

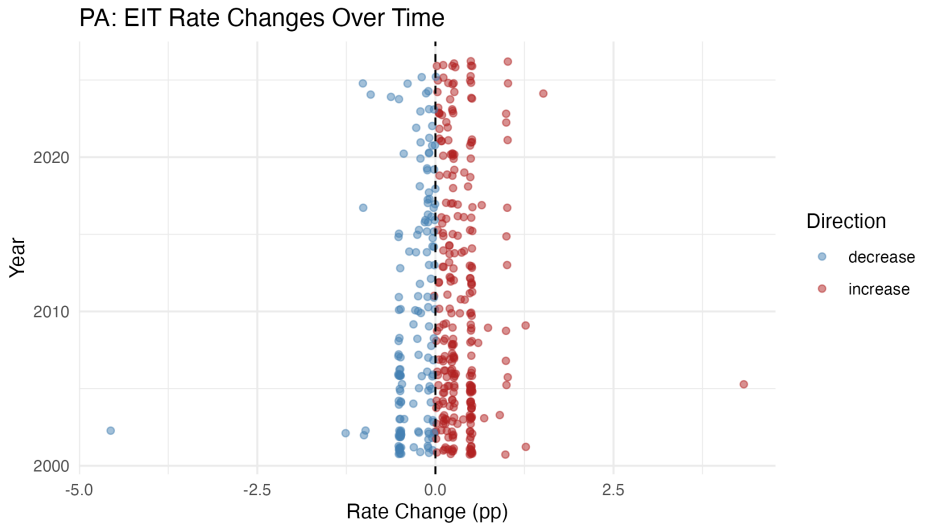


Figure 2: Pennsylvania: Distribution of EIT Rate Changes Over Time

5 Empirical Strategy

All specifications are estimated at the muni × year level with muni and year fixed effects, with standard errors clustered at the muni level. The muni fixed effects absorb time-invariant differences across jurisdictions (size, geography, baseline rates); the year fixed effects absorb statewide shocks (business cycles, LODES coverage changes, statewide policy). Identification comes from within-muni rate variation over time.

5.1 Specifications

Q2 (headline TWFE). The headline test is a continuous-rate two-way fixed-effects regression in two parallel forms—a bundled total-stack regressor and a decomposition into the muni and SD components:

$$\text{(A) Total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{3}$$

$$\text{(B) Decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{4}$$

where $\tau_{it}^{\text{home}} = \tau_{it}^{\text{res,muni}} + \tau_{it}^{\text{SD}}$ is the home-side rate stack faced by residents of muni i, $Y_{it} \in \{\text{share_live_in}, \log(\text{resident workers}), \log(\text{total jobs})\}$, and $\mathbf{X}_{it}$ is the control vector: log spending on police, streets, debt service, and state aid, plus revenue shares (EIT, real estate, intergov as fractions of total revenue). We avoid including expenditure categories that are downstream of the rate itself (community development in particular grows with rate increases and would constitute a bad control).

Sun–Abraham aggregate ATT. For each outcome we also report the Sun and Abraham (2021) interaction-weighted aggregate ATT, which treats the rate increase as a binary cohort event with never-changed munis as controls:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{5}$$

where D_{it}^{ℓ} indicates muni i being ℓ years from its first muni-rate increase. Pre-event coefficients test parallel trends. The aggregate ATT is the IW-weighted average of post-event coefficients.

DCDH continuous treatment. For staggered designs with a continuous dose, we estimate the de Chaisemartin et al. (2024) average causal response (ACR) per pp, which is robust to the negative-weighting issue in TWFE (Goodman-Bacon, 2021). Implementation: `did_multiplegt_dyn` with continuous treatment, 5 effect periods, 5 placebo periods, and the same fiscal controls as the TWFE specifications. Bootstrap SEs recommended for continuous treatment but not used in the headline table due to runtime cost; results in the paper use analytical SEs, which are conservative (the package warns).

Act 32 structural break. Prediction 1 implies the sorting response should differ before and after the 2012 consolidation of tax collection. We interact the rate with pre- and post-Act

32 indicators:

$$Y_{it} = \alpha_i + \gamma_t + \beta_m^{\text{pre}} \tau_{it}^{\text{res,muni}} \mathbf{1}[t < 2012] + \beta_m^{\text{post}} \tau_{it}^{\text{res,muni}} \mathbf{1}[t \geq 2012] + \beta_{sd}^{\text{pre}} \tau_{it}^{\text{SD}} \mathbf{1}[t < 2012] + \beta_{sd}^{\text{post}} \tau_{it}^{\text{SD}} \mathbf{1}[t \geq 2012] + \mathbf{X} \tag{6}$$

Under Prediction 1, $\beta_m^{\text{pre}} \approx 0$ and $\beta_m^{\text{post}} < 0$. The SD coefficient should be similar in both periods because school-district withholding (administered through the state PIT return) was already in force pre-Act 32.

Q3 clean groups (asymmetry). We partition munis by muni-rate event type: inc-only (ever raised, never decreased; 97 munis), dec-only (24), never-changed (1,264). Flip-floppers (37% of changers) are dropped. The estimator is Equation (7): muni rate interacted with group, plus SD rate as a single coefficient (controlling for muni rate). Asymmetry holds if $|\beta_{\text{inc}}^m| > |\beta_{\text{dec}}^m|$.

6 Results

This section presents the muni-level estimates. Q2 establishes the headline per-pp effect of muni and school district rates on `share_live_in` and the count outcomes. Q3 estimates the asymmetric response by restricting to clean groups (muni-rate increasers vs. never-changers, decreasers vs. never-changers, dropping flip-floppers) and using the triple-interaction specification.

6.1 Q2: Do Rate Increases Cause Sorting?

Table 2 reports the headline TWFE estimates of muni and school-district rate effects on three outcomes: `share_live_in` and the log counts of resident workers and total jobs. All specifications include muni and year fixed effects and the controls listed in Section 5.1: log expenditure on police, streets, debt, and state aid, plus revenue shares (EIT, real estate tax, intergovernmental as fractions of total revenue).

Table 2: Q2: TWFE estimates with full controls and revenue shares

	share_live_in	log(Resident)	log(Total Jobs)
Total stack			
τ^{home}	-0.004	-0.049	-0.092^{**}
	(0.003)	(0.040)	(0.040)
Decomposition			
$\tau^{\text{res,muni}}$	-0.008^{*}	-0.113^{*}	-0.119^{**}
	(0.004)	(0.067)	(0.059)
τ^{SD}	$+0.009$	$+0.147^{**}$	-0.009
	(0.007)	(0.059)	(0.046)
Sun–Abraham aggregate ATT (binary cohort)			
SA ATT	-0.003	-0.031	-0.030
	(0.003)	(0.022)	(0.026)
R^2	0.898	0.972	0.972
N	13,199	13,199	13,199

Notes: Sample is PA municipalities, 2002–2019, with full fiscal controls available. $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$ is the home-side rate stack. Decomposition enters the muni and SD components separately. Sun–Abraham rows use binary cohorts (first muni rate increase) with never-changed munis as controls; reported coefficient is the IW-weighted aggregate ATT. Cluster-robust SEs by muni in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A 1 pp increase in the muni resident rate lowers `share_live_in` by 0.8 share-points and resident worker counts by 11%, controlling for the full set of fiscal and revenue share controls. The total-stack regressor is essentially null on share_live_in and resident workers because the muni and SD components move outcomes in opposite directions and partially cancel: the SD coefficient on log(Resident) is $+0.147^{**}$, indicating a 15% rise in resident workers per pp of SD rate. The decomposition is therefore not optional—bundling the home rate hides both channels.

The Sun–Abraham aggregate ATT (binary cohort) is consistent in sign but smaller in magnitude than the per-pp TWFE coefficient because it averages across event times (including pre-effect-buildup years) and treats the rate change as a binary event rather than a continuous dose. Figure 3 shows the dynamics: effects build over 5–8 years, with cleaner identification in the muni cohort than the SD cohort.

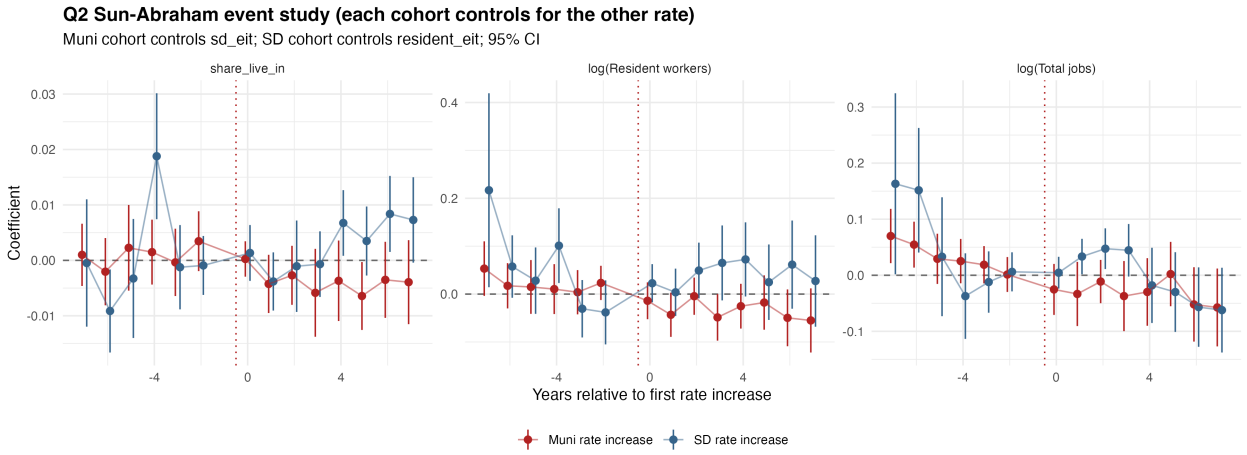


Figure 3: Q2 Sun–Abraham event study, muni vs. SD cohorts decomposed. Each cohort's spec controls for the other rate; pair fiscal controls + revenue shares. 95% CI.

The de Chaisemartin et al. (2024) continuous-treatment estimator with fiscal controls gives an average causal response of −0.009 on share_live_in, −0.078 on log(Resident), and −0.067 on log(Total Jobs) per pp of muni rate. With analytical SEs (the package warns these are conservative for continuous treatment without bootstrap), the share_live_in coefficient does not clear conventional significance but the magnitude tracks the TWFE estimate.

Act 32 structural break. The muni-level sorting response is concentrated entirely in the post-Act 32 period (Table 3).

Table 3: Act 32 structural break: rate effect split pre/post 2012

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Pre	-0.004	-0.056	-0.078
	(0.004)	(0.050)	(0.047)
$\tau^{\text{res,muni}} \times$ Post	-0.011^{**}	-0.151^{**}	-0.144^{**}
	(0.005)	(0.066)	(0.058)
$\tau^{\text{SD}} \times$ Pre	$+0.008$	$+0.147^{**}$	$+0.014$
	(0.007)	(0.061)	(0.048)
$\tau^{\text{SD}} \times$ Post	$+0.013^{*}$	$+0.170^{**}$	-0.032
	(0.007)	(0.071)	(0.056)
R^2	0.899	0.972	0.972
N	13,199	13,199	13,199

Notes: Pre-period = 2002–2011, post = 2012–2019. Same controls and FE as Table 2. Pre-2012 the muni rate coefficient is statistically zero on every outcome; post-2012 it is negative and significant. The SD effect on resident workers is present in both periods, slightly stronger post-2012. Workers respond to taxes that are actually being collected. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The pre-2012 muni coefficient on `share_live_in` is -0.004 (SE 0.004), statistically indistinguishable from zero. Post-2012 it is -0.011^{**}. The same pattern holds across log(Resident) and log(Total Jobs): pre-period coefficients are smaller and not significant; post-period coefficients are larger and significant at the 5% level. This is direct evidence of Prediction 1: workers respond to taxes they actually pay, and the same statutory rate generates a much larger behavioral response when employer withholding is binding. The SD coefficient on log(Resident) is positive and significant in both periods because Pennsylvania's school-district withholding (administered through the state PIT return) was already in force pre-Act 32.

6.2 Q3: Are Rate Changes Asymmetric?

The Q2 specifications pool all rate variation. To test asymmetry cleanly, we partition munis into three groups by the type of rate events they experienced over 2002–2019:

- **Inc-only** (97 munis): ever raised the muni rate, never decreased it.
- **Dec-only** (24 munis): ever decreased the muni rate, never raised it.
- **Never-changed (NC)** (1,264 munis): muni rate flat throughout the panel.

Munis that both raised and decreased their rate at some point during the panel ("flip-floppers", ∼37% of changers) are dropped. This partition isolates within-group identification: the inc-only group's muni-rate variation is unambiguously upward; dec-only is unambiguously downward; never-changed contributes no muni-rate variation but contributes year fixed-effect identification.

The estimating equation interacts the continuous muni rate with the group indicator and includes the SD rate as a single coefficient (controlling for the muni rate, what is the SD effect?):

$$Y_{it} = \alpha_i + \gamma_t + \beta_g^m \tau_{it}^{\text{res,muni}} \cdot G_i + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{7}$$

where $G_i \in \{\text{nc, inc, dec}\}$. The interaction $\tau^{\text{res,muni}} \times$ NC is dropped due to collinearity (no within-variation in muni rate for never-changers).

Table 4: Q3: Clean groups (inc-only / dec-only vs. never-changed)

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.020^{***}	-0.256^{***}	-0.213^{***}
	(0.006)	(0.046)	(0.047)
$\tau^{\text{res,muni}} \times$ Dec	-0.025	-0.224	-0.208
	(0.022)	(0.176)	(0.162)
τ^{SD}	$+0.007$	$+0.139^{**}$	-0.026
	(0.006)	(0.071)	(0.052)
R^2	0.894	0.970	0.971
N	12,745	12,745	12,745

Notes: Same controls and FE as Table 2. Inc-only = 97 munis, Dec-only = 24, Never-changed = 1,264. Flip-floppers dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The muni-rate effect among inc-only munis is -2.0 share-points per pp on `share_live_in`, -26% on resident workers, and -21% on total jobs—all significant at the 1% level. These magnitudes are roughly twice the pooled Q2 estimates because the inc-only sample isolates clean upward variation without dilution from flip-floppers or stable-rate cities. The dec-only point estimates are similar in sign and magnitude but statistically imprecise because the group has only 24 munis. The data are consistent with sticky moving costs (rate cuts do not detectably reverse the displacement) but cannot definitively reject symmetric responses given the small dec-only sample.

The SD coefficient is $+14\%^{**}$ on log(Resident), close to its Q2 counterpart and unaffected by the group structure. This is the opposite-sign pattern: muni rate repels residents while SD rate attracts them.

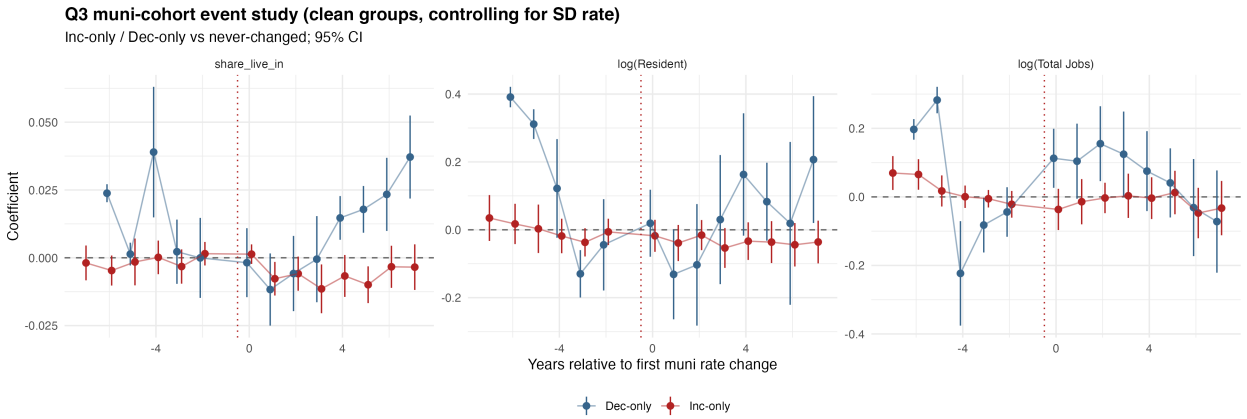


Figure 4: Q3 Sun–Abraham event study, clean groups. Inc-only and Dec-only munis vs. never-changed controls; muni-rate cohort. Each spec controls for the SD rate. 95% CI.

Substitution between muni and SD rate changes. The clean-groups partition reveals a fiscal-substitution pattern: muni-rate-decreasers are roughly five times more likely to have simultaneous SD rate increases than the never-changed group (50% vs 10%), while muni-rate-increasers are mostly in stable-SD regions (78% SD never-changed). This is consistent with the institutional design of Act 1 of 2006, which created the SD rate as an alternative to muni-side revenue. Including the SD rate as a control is therefore not optional: omitting it would conflate SD response with muni-rate identification in the dec-only group.

6.3 Heterogeneity

A standard prediction of the tax migration literature is that high-income workers should be more responsive to tax changes (Kleven et al., 2020; Young et al., 2016). LODES provides two earnings segments (SE01: \$1,250/month or less; SE03: \$3,333/month or more) and three age segments (SA01: 29 or younger; SA02: 30–54; SA03: 55 or older). We estimate the PA dose-response specification separately for each segment.

The earnings and age segments respond nearly identically: the dose-response coefficient is approximately -0.009^{**} for both low-earnings (SE01) and high-earnings (SE03) workers, and similar across all three age groups. We find no evidence that higher-earning or older

workers are more responsive to local EIT rate changes.

This null heterogeneity result likely reflects the coarseness of the LODES earnings categories. The SE03 threshold ($40,000 per year) captures middle-income workers, not the high earners who drive state-level tax migration results. The local income tax rates in our sample (0.5–3.0%) generate modest dollar differences at these income levels: a 1 percentage point rate increase costs a $40,000 earner $400 per year, unlikely to produce differential migration responses by earnings category. True heterogeneity testing would require linked employer-employee data with continuous earnings, which LODES does not provide.

7 Robustness

We assess robustness along four dimensions: fiscal controls, Philadelphia exclusion, pre-treatment balance, and the Act 32 structural break across the municipal and fiscal outcomes.

Table 5: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	−0.0128**	−0.0104**	−0.0167***			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				−0.0090***	−0.0081***	−0.0098***
				(0.0027)	(0.0025)	(0.0029)
N	27,977	14,644	27,957	3,258	2,336	3,238
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7.1 Act 32 Structural Break Across All Outcomes

If Act 32 enforcement is the mechanism behind the municipal-level sorting result (Table 3), the same pre/post-2012 pattern should appear across our other municipal outcomes. We test this by estimating Equation 6 for resident worker counts, total jobs, and fiscal revenues.

Municipal sorting and school district decomposition. Table 6 reports the pre- vs post-Act 32 split for municipal-level outcomes. The municipal EIT effect is a precise zero before 2012 and significant after 2012 across all outcomes. The school district decomposition reveals that the SD amenity effect on resident workers is positive in both periods but roughly doubles after Act 32 (+0.103** to +0.196***), consistent with consolidated withholding making all EIT variation more salient.

Table 6: Act 32 Robustness: Stage 1 Pre/Post-2012 Split

	(1) share_live_in	(2) log(Resident)	(3) log(Total Jobs)
Panel A: Municipal EIT only			
$\tau^m \times$ Pre-2012	-0.002 (0.004)	-0.010 (0.041)	-0.032 (0.033)
$\tau^m \times$ Post-2012	-0.018^{***} (0.005)	-0.198^{***} (0.057)	-0.127^{***} (0.043)
Panel B: Municipal + School District EIT			
$\tau^m \times$ Pre-2012	-0.002 (0.004)	-0.044 (0.050)	
$\tau^m \times$ Post-2012	-0.017^{***} (0.005)	-0.195^{***} (0.059)	
$\tau^{sd} \times$ Pre-2012	-0.001 (0.006)	0.103^{**} (0.044)	
$\tau^{sd} \times$ Post-2012	0.007 (0.007)	0.196^{***} (0.047)	
Panel C: Total EIT (municipal + school district)			
$\tau^{\text{total}} \times$ Pre-2012	-0.006 (0.004)		
$\tau^{\text{total}} \times$ Post-2012	-0.005 (0.004)		
N	27,977	24,448	24,448
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Each cell is from a separate regression. Panel A: municipal EIT rate interacted with pre/post-2012 indicators. Panel B: both municipal and SD rates interacted with period indicators. Panel C: combined total EIT rate. The municipal effect is entirely post-2012; the SD amenity effect is present in both periods but stronger post-2012. Total EIT is null in both periods because the opposing municipal and SD effects cancel. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fiscal outcomes. If Act 32 made the EIT binding, the same statutory rate should generate more revenue after 2012. Table 7 confirms this. Holding the EIT rate constant, a one percentage point rate generates 5.1% more EIT revenue post-Act 32 ($p < 0.05$). The effect is specific to the EIT: property tax revenue and total tax revenue show no structural break at 2012, ruling out a general fiscal trend. Total revenue shows a *negative* post-2012 interaction (-0.094^{***}), consistent with the pre-2012 positive coefficient reflecting correlated

fiscal expansion rather than an enforcement effect.

Table 7: Act 32 Fiscal Structural Break: EIT Rate → Revenue

	(1)	(2)	(3)	(4)
	log(EIT Rev)	log(Total Tax)	log(Total Rev)	log(RE Tax)
τ (baseline)	0.650^{***}	0.296^{***}	0.120^{**}	-0.082
	(0.213)	(0.101)	(0.053)	(0.068)
$\tau\times$ Post-2012	0.039	0.002	-0.091^{***}	0.031
	(0.024)	(0.019)	(0.025)	(0.036)
N	43,407	43,626	43,643	42,750
Fixed Effects	Municipality, Year			
Clusters	Municipality			

Notes: PA DCED Annual Financial Reports, 2002–2019. Each column is a separate regression of log revenue on the municipal EIT rate and its interaction with a post-2012 indicator, with municipality and year fixed effects. The baseline coefficient captures the rate–revenue relationship across the full panel; the interaction captures the additional enforcement effect of Act 32. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Summary. The Act 32 structural break is consistent across the municipal-level outcomes we examine: sorting, resident worker counts, total jobs, and EIT revenue. The municipal EIT effect is a precise zero before 2012 and significant after 2012, while school district amenity effects are present in both periods and strengthen post-2012. Property tax revenue and total tax revenue show no structural break, ruling out a general fiscal trend.

8 Discussion

The results confirm the Tiebout hypothesis at the local level—workers do sort across municipal boundaries in response to income tax rate changes—but the magnitude and mechanism depart from the canonical prediction in several ways.

The response is real but small. A 1 percentage point municipal EIT rate increase reduces the locally-resident worker share by roughly 1 point and log resident workers by 13–14%. These are economically meaningful: a typical PA municipality with 700 resident workers would lose approximately 90–100 over 8 years. But the behavioral response is far too small to erode the tax base. EIT revenue rises 67% per percentage point of rate increase, total tax revenue rises 30%, and population and property values are unaffected. Municipalities that raise rates are on the correct side of the revenue hill (Haughwout et al., 2004).

Enforcement is the mechanism, not tax salience. The Act 32 structural break is the paper's strongest identification result. Before 2012, when nonresident EIT collection was fragmented and largely unenforced, the same statutory rate produced no detectable sorting. After 2012, when consolidated employer withholding made the tax binding, the effect is large and precisely estimated. This pattern holds across all municipal outcomes: sorting, resident worker counts, total jobs, and EIT revenue. The implication is that the relevant margin is not tax salience or information—workers know their tax rate—but whether the tax is actually collected. Prior studies of local tax competition that do not account for enforcement quality may underestimate the behavioral response.

Asymmetry implies sticky residential choices. Rate increases push residents out; rate decreases do not bring them back. This asymmetry is consistent across estimators and all outcome measures. The most natural interpretation is that residential relocation involves fixed costs (moving expenses, social ties, school enrollment) that make the decision irreversible at the margin. A rate increase that pushes a worker past the threshold for moving cannot be reversed by a rate decrease of equal magnitude, because the worker has already incurred the fixed cost and reoptimized. This has policy implications: a municipality that raises and then lowers its rate will not recover the residents it lost.

School district rates operate through a different channel. The opposite sign on municipal and school district EIT rates is among the paper's most striking findings. Municipal rate increases fund general operations; school district rate increases were authorized under Act 1 of 2006 specifically for property tax relief. The Q3 decomposition shows municipal rate increases repel residents while SD rate increases attract them, with the SD attraction effect roughly offsetting municipal rate repulsion in high-SD areas. Whether the attraction reflects actual property tax relief, a school quality signal, or unobserved amenity changes correlated with SD rate increases remains an open question. This pattern explains why studies using the total EIT rate find null results: they conflate two taxes with opposite behavioral effects.

Uniform response across earnings challenges selective migration. The tax migration literature emphasizes that high earners are more responsive to tax differentials (Kleven et al., 2020; Young et al., 2016). We find no differential response by LODES earnings category. While the coarseness of LODES earnings segments limits the test (the highest category captures workers earning $40,000+, not top earners), the uniform response is consistent with local income tax sorting operating through a different mechanism than state-level tax migration: at the dollar amounts involved (a few hundred dollars per year), the relocation decision is driven by cumulative quality-of-life considerations rather than tax optimization by high-net-worth individuals.

Limitations. Three limitations merit acknowledgment. First, LODES employment data are subject to noise infusion at the census block level; while aggregation to the municipality level substantially reduces measurement error, it introduces a mechanical tendency toward attenuation bias. Second, our design cannot distinguish tax-motivated sorting from sorting driven by correlated amenity changes (e.g., a municipality raises rates and simultaneously cuts services). The fiscal controls and pre-treatment balance tests mitigate this concern, but we cannot rule out all confounds. Third, the LODES reference date (April 1) captures a snapshot of employment, not the full year; seasonal or temporary workers may introduce

noise in municipalities with volatile employment.

9 Conclusion

This paper provides causal evidence that local income tax rate changes cause residential sorting across municipal boundaries. Using LEHD Origin-Destination data for Pennsylvania (2002–2019), I show that rate increases reduce the locally-resident share of a city's workforce, that the response is asymmetric (increases push residents out but decreases do not bring them back), and that the sorting is entirely concentrated after Pennsylvania's Act 32 enforcement consolidation in 2012.

The decomposition between municipal and school district EIT rates reveals that they operate through opposite channels: municipal rate increases repel residents while school district rate increases attract them, with the SD attraction roughly offsetting municipal rate repulsion in high-SD-rate areas. Despite the behavioral response, rate increases are fiscally productive: revenue gains dominate the modest base erosion from sorting.

These findings have implications for municipal fiscal policy. First, local income tax rate increases "work" in the narrow fiscal sense: they raise revenue without measurably reducing population or property values. Second, the enforcement mechanism matters as much as the statutory rate; municipalities considering rate increases should ensure collection capacity. Third, the school district decomposition suggests that not all local tax increases are alike: SD rate increases are associated with residential attraction rather than repulsion, though whether this reflects property tax relief, school quality signaling, or correlated amenities cannot be fully disentangled with the available fiscal data.

The LODES-based approach developed here—linking worker residential choices to municipal tax variation at annual frequency—can be applied to other local fiscal instruments (property taxes, sales taxes, fees) and to related questions about the spatial incidence of local policy.

Acknowledgments

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A Pre-Treatment Balance

Table 8: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
Total Debt ($)	28.6M	3.6M	0.089	
Market Value RE ($)	8.2M	2.1M	0.211	523 / 8,774
Total Index Crime	985	133	0.124	816 / 18,322
Building Permits	19.3	7.7	0.232	644 / 25,066
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. Normalized differences < 0.15 for all fiscal variables except market value of real estate (0.211, limited pre-2012 coverage) and building permits (0.232, treated municipalities are larger and issue more permits in levels). Sources: PA DCED Annual Financial Reports, FBI UCR Offenses Known (93.5% agency match rate), Census Building Permits Survey (93.3% MCD match rate).

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on eleven observables—six fiscal variables from the PA DCED Annual Financial Reports, real estate market values, total employment from LODES, Part I index crime from the FBI Uniform Crime Reports, and housing units permitted from the Census Building Permits Survey—using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 5 with municipality and year fixed effects, clustering

standard errors at the municipality level. Table 9 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for nine of eleven observables: total revenue (max $|t| = 1.57$), EIT revenue (1.40), RE tax revenue (1.69), streets expenditure (0.70), population (1.09), total debt (1.36), total jobs from LODES (0.83), index crime (0.68), and building permits (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level. The jobs, crime, and permits results are particularly important: they confirm that treated and control municipalities were on parallel trajectories in employment, public safety, and housing development before the rate change, ruling out differential economic trends as confounds.

Police expenditure is an exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase. This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table 5), confirming that this pre-trend does not confound the estimated sorting response.

Market value of real estate also shows a pre-trend at $\ell = -3$ ($t = -2.59$), but this variable has limited coverage (pre-2012 only, $N = 9{,}338$) and the pre-trend may reflect composition effects in the truncated sample rather than a genuine violation of parallel trends.

Table 9: Pre-Trend Tests: Sun-Abraham on Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.57	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.40	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(RE Tax Revenue)	−0.025	−0.060	−0.011	−0.015	1.69	41,683
	(0.044)	(0.036)	(0.029)	(0.017)		
log(Police Exp)	−0.085***	−0.068***	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.70	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		
log(Market Val RE)			−0.403***	−0.026	2.59	9,338
			(0.156)	(0.016)		
log(Total Jobs)	0.006	0.018	−0.001	−0.005	0.83	24,557
	(0.023)	(0.022)	(0.015)	(0.013)		
log(Index Crime)	−0.125	−0.094	−0.084	−0.031	0.65	20,613
	(0.194)	(0.151)	(0.236)	(0.155)		
log(Bldg Permits)	−0.030	−0.057	0.084	0.171	1.35	27,515
	(0.222)	(0.157)	(0.147)	(0.126)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure (lower spending at $\ell = -5$ and $\ell = -4$) and market value of real estate (limited to pre-2012 data, $N = 9{,}338$). Total jobs from LODES; index crime from FBI UCR Offenses Known (93.5% agency-to-municipality match);